EKS Spot index-calculus curve-size validation

2026-10-26 · public synthetic, one target per curve

Four independent `c8a.2xlarge` Spot Jobs ran the frozen `ecbench.spec/v1` panel at 20, 24, 28 and 32 bits. Each job used five measured rounds and one warmup, with `ic.pipeline`, matched `rho.negation`, and a rho A/A control. All 72 records were verified, all four native audits passed, and each job earned 18/18 L2 records. Each audit included 15 exact replays. The repository `README.md` and `n*/` receipts provide exact specs, session records, audit reports, source/image identity, Job/pod IDs, and S3 prefixes.

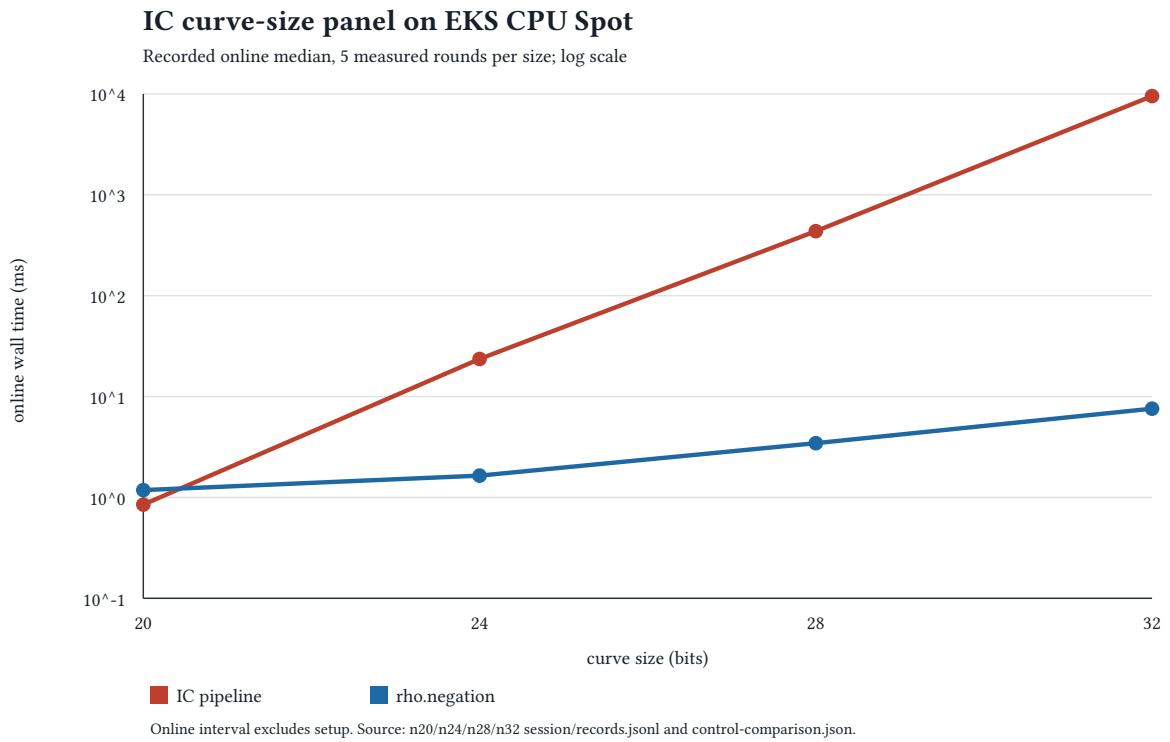

Figure 1: Recorded online medians on a logarithmic scale. Five measured rounds per size. Source: `n*/session/records.jsonl` and `n*/control-comparison.json`.

The recorded IC online medians were 0.849690, 23.555580, 435.896655, and 9520.577446 ms for 20, 24, 28, and 32 bits respectively. Matched rho medians were 1.183083, 1.646459, 3.453848, and 7.591005 ms. The corresponding IC total GAE medians were 8529, 165732, 2516038, and 31077180; rho medians were 1016, 4510, 19263, and 49663. These values are descriptive diagnostics. Online timing excludes setup; no end-to-end `S`, external comparison, or precomputation pricing band was established. The 28 and 32-bit A/A confidence intervals are wide. This report makes no speedup or security claim.

The checked image digest is `sha256:67eea58f5d4a7a8882b55f089c6063bb49fad58d40260ee9b0ceb349fc0e4581`; the native executable SHA-256 is `583e62f0836eae43330c4b36bbe3d145dacb01de3132436c71f6924445c6a5a9`. The new Actions dispatch and CRIU restore path require a separate post-merge validation; the existing Spot panel did not exercise worker interruption.